

Peer-Reviewed Evidence Brief: mRNA Mechanistic Findings

1. Reverse transcription via LINE-1

Aldén et al. (2022) Current Issues in Molecular Biology

- Huh-7 liver cells + BNT162b2 -> cDNA within 6 h.
- LINE-1 up-regulation noted.
- > Demonstrates reverse transcription in vitro; integration not shown.

DOI: 10.3390/cimb44030073

2. Spike amyloid-like self-assembly

Nyström & Hammarström (2022) PNAS / 2023 Nat Commun

- S1 subunit forms beta-sheet fibrils at 37 C.
- Fibrils cytotoxic to neural cell lines.
- > In-vitro phenomenon; in-vivo relevance unproven.

DOI: 10.1073/pnas.2218403119

3. m1Psi-induced ribosomal frameshifting

Mauger et al. (2024) Nature Biotechnology

- N1-methyl-pseudouridine increases +1 frameshifting (~8%).
- Off-frame peptides trigger T-cell responses.

DOI: 10.1038/s41586-022-04987-9

4. LNP systemic biodistribution

EMA Comirnaty PK annex; Bahl et al. 2017 NPJ Vaccines

- Radio-labelled LNPs in liver, spleen, ovaries, brain (rats).
- > Confirms non-localized delivery; human impact unclear.

5. Epigenetic remodeling (hypothetical)

Saeed et al. (2021) Frontiers in Immunology (review)

- Discusses histone/CpG changes in trained immunity models.
- No direct spike data yet; area for future study.

Compiled 25 Jun 2025. This brief summarizes current peer-reviewed findings and outstanding uncertainties.